

Daniel P. Keliher

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Interests	I am interested in arithmetic statistics and number theory , including field counting, ranks of elliptic curves, Selmer groups, prime number races, special values of L -functions, explicit estimates, and integer partitions. I am also interested in the interplay of these areas with computational mathematics, probability, statistics, and optimal transport. Previously, I worked in computational biology.	
Employment	<i>Massachusetts Institute of Technology</i> , Cambridge, MA Concourse Program Lecturer	July 2024-July 2026
	<i>University of Georgia</i> , Athens, GA Limited Term Assistant Professor	Aug. 2022-June 2024
Education	<i>Tufts University</i> , Medford, MA Ph.D in Mathematics Advisor: Robert Lemke Oliver	May 2022
	<i>Tufts University</i> , Medford, MA M.S. in Mathematics	May 2019
	<i>Brown University</i> , Providence, RI Sc.B in Mathematics	May 2017
Funding	MIT Education Innovation Fund Grant. \$43,673. (Co-PI with Erica Zimmer)	June 2025 - June 2026
Papers/ Preprints	11. <i>Entropy of some distributions defined with the Ihara zeta function</i>, with Annie Xu. In preparation. 10. <i>Large Deviation Principles for Abelian Monoids</i>, with Sun Woo Park. Electronic Communications in Probability, 2026. (arXiv) 9. <i>Rank Growth of Elliptic Curves in S_3 Cubic Extensions</i>, with Sun Woo Park. Submitted, 2025 (arXiv) 8. <i>Exponents of Jacobians and relative class groups</i>, with Borys Kadets. Proceedings of the American Mathematical Society, 2026. (arXiv) 7. <i>Rank Growth of Elliptic Curves in S_4 and A_4 Quartic Extensions of the Rationals</i>. Pacific Journal of Mathematics, 2024. (arXiv) 6. <i>On the Constants in Mertens' Theorems for Primes in Arithmetic Progressions</i>, with Ethan Lee. Proceedings of the Integers Conference 2023, 2024. (arXiv) 5. <i>Exceptional Biases Over Function Fields</i>, with Alexandre Bailleul, Lucile Devin, and Wanlin Li. Journal of the London Mathematical Society, 2024. (arXiv) 4. <i>Optimal Transport for Some Symmetric Multidimensional Partitions</i>, with Daniel Adu. Discrete Applied Mathematics, 2023. (arXiv)	

3. *Enumerating D_4 Quartics and a Galois Group Bias Over Function Fields*. *Journal de Théorie des Nombres de Bordeaux*, 2022. (arXiv)
2. *Mertens' Theorem for Chebotarev Sets*, with Santiago Arango-Piñeros and Christopher Keyes. *International Journal of Number Theory*, 2022. (arXiv)
1. *Comparing the Density of D_4 and S_4 Quartic Extensions of Number Fields*, with Matthew Friedrichsen. *Proceedings of the American Mathematical Society*, 2021. (arXiv)

**Papers
(Comp. Bio.)**

5. *Integrating molecular profiles into clinical frameworks through the Molecular Oncology Almanac to prospectively guide precision oncology*. Brendan Reardon, Nathanael Moore, Nicholas S Moore, et al. *Nature Cancer*. 2021.
4. *Immunogenomic analyses associate immunological alterations with mismatch repair defects in prostate cancer*. Daniel Nava Rodrigues, Pasquale Rescigno, David Liu, et al. *Journal of Clinical Investigation*, 2018.
3. *Genomic correlates of response to immune checkpoint blockade in microsatellite-stable solid tumors*. Diana Miao, Claire A. Margolis, Natalie I. Vokes, et al. *Nature Genetics*, 2018.
2. *Mutational patterns in chemotherapy resistant muscle-invasive bladder cancer*. David Liu, Philip Abbosh, Daniel Keliher, et al. *Nature Communications*, 2017.
1. *Subclonal mutational heterogeneity and survival in cisplatin-resistant muscle-invasive bladder cancer*. David Liu, Philip Abbosh, Daniel Keliher, et al. *Journal of Clinical Oncology*, 2017.

Book Draft

From Circles to Cubics: A Tour of Elementary Algebraic Geometry, with Borys Kadets. (Full draft available on request, 250+ pages)

**Teaching
Experience**

Fall 2023: Awarded Writing Intensive Program funding (UGA)
Spring 2020: Mary Glaser Distinguished Teaching Prize (Tufts)

Term	Course Name	Number	Inst.	
Spring 2026	Differential Equations	CC1803	MIT	
Spring 2026	Multivariable Calculus	CC1802	MIT	
Fall 2025	Multivariable Calculus	CC1802	MIT	
Summer 2025	Linear Algebra	MATH0070	Tufts	
Spring 2025	Independent Study	CC0220	MIT	(two sections)
Spring 2025	Multivariable Calculus	CC1802	MIT	
Spring 2025	Differential Equations	CC1803	MIT	
Fall 2024	Multivariable Calculus	CC1802	MIT	
Fall 2024	Calculus Recitation	CC1801	MIT	(recitation only)
Spring 2024	Calculus I	MATH2250	UGA	(two sections)
Fall 2023	Cryptography	MATH6450	UGA	(graduate)
Fall 2023	Cryptography	MATH4450	UGA	(concurrent w/ 6450)
Spring 2023	Calculus II	MATH2260	UGA	
Fall 2022	Calculus I	MATH2250	UGA	(two sections)
Spring 2022	Teaching Seminar II		Tufts	(as Dept. Head TA)
Spring 2022	Calculus Recitation	MATH0032	Tufts	(recitation only)
Fall 2021	Teaching Seminar I		Tufts	(as Dept. Head TA)
Summer 2021	Math of Social Choice	MATH0019	Tufts	
Spring 2021	Math for Social Science	MTH171	Salve	
Summer 2020	Introductory Statistics	MATH0021	Tufts	
Spring 2020	Introductory Statistics	MATH0021	Tufts	

Invited Talks

- Colby College Mathematics Colloquium, April 2026
- Middlebury College Mathematics and Statistics Seminar, January 2026
- MIT Digital Humanities + Data Seminar, October 2025
- Five College Number Theory Seminar (at Amherst College), February 2025
- James Madison University Math Dept. Colloquium, February 2024
- Bentley University Mathematical Sciences Seminar, January 2024
- Institute for Defense Analyses, Center for Communications Research (Princeton) Colloquium, January 2024
- University of South Carolina Algebra, Geometry, and Number Theory Seminar, December 2023
- AMS Fall Southeastern Sectional Special Session “Number Theory and Friends”, University of South Alabama, October 2023
- AMS Central Sectional Special Session on Arithmetic Statistics, University of Cincinnati, April 2023
- Joint Math Meetings, Special Session on Arithmetic Statistics, January 2023
- Emory University Algebra and Number Theory Seminar, November 2022
- University of Georgia Number Theory Seminar, October 2022
- University of Georgia Oberseminar, August 2022
- AMS Virtual Spring Eastern Sectional Meeting, March 2022
- Duke University Number Theory Seminar, February 2022
- Montréal Online Biweekly Inter-University Seminar on Analytic Number Theory, November, 2021

Contributed Talks

- Southern Regional Number Theory Conference, Louisiana State University, March 2024
- INTEGERS 2023, University of Georgia, May 2023
- Southern Regional Number Theory Conference, Louisiana State University, March 2023
- Upstate Number Theory Conference, Union College, October 2021
- Graduate Student Conference in Algebra, Geometry, and Topology, Temple University, May 2021
- Tufts Graduate Student Summer Speaker Series, Virtual Series, Tufts University, June 2020
- Modular Forms, Arithmetic and Women in Mathematic, Emory University, November 2019
- Maine-Québec Number Theory Conference, University of Maine, October 2019
- Graduate Student Conference in Algebra, Geometry, and Topology, Temple University, May 2019
- Palmetto Number Theory Series, University of South Carolina, December 2018
- Tufts University Math Graduate Student Seminar (4 Talks)

Service	• Refereed papers at the following journals: ◦ European Journal of Mathematics ◦ Proceedings of the American Mathematical Society ◦ Research Directions In Number Theory ◦ Mathematische Nachrichten • MIT First-year advisor. ◦ 6 advisees 2025-26 ◦ 8 advisees 2024-25 • Co-organizer, special session on “Counting and Asymptotics in Number Theory” at the AMS Spring Eastern Sectional Meeting, April 2025 • Conference organizer, Palmetto Area Number Theory Series (PANTS XXXVII), December 2023 • University of Georgia Calculus 1 Exam Committee, Fall 2022 • Mentor, Spelman-Morehouse Joint Direct Reading Program, Fall 2022 • Co-Organizer, Tufts Algebra/Geometry/Number Theory Seminar, Fall 2021 • Organizer, Tufts Graduate Student Seminar, Spring 2018-Fall 2021 • Tufts University Directed Reading Program in Mathematics: ◦ Co-organizer for the DRP, Fall 2019 - Spring 2022 ◦ Mentor for a DRP project on Elliptic Curves, Fall 2019 ◦ Mentor for a DRP project on Information Theory, Spring 2020 • Organizer, Reading Seminar on Arithmetic Geometry, Spring 2020 (with participants from Tufts, Emory, and Georgia Tech.) • Tufts University Organization for Graduate Students in Mathematics (OGSM) ◦ First-Year Representative, AY 2017-2018 ◦ Second-Year Representative, AY 2018-2019
Professional Memberships	• American Mathematical Society • The National Association of Mathematicians
Computer Skills	R, Python, L ^A T _E X, Mathematica, Sage, Magma